| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/31/2006 |
| **PI Name:** | Pishko, Michael V Ph.D. | | |
| **Project Title:** | Microfabricated Optical Biosensor Arrays for Air Quality Monitoring | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | mpishko@engr.psu.edu | **Fax:** | FY 814-865-7846 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 814-863-4810 |
| **Organization Name:** | The Pennsylvania State University | | |
| **PI Address 1:** | Department of Chemical Engineering | | |
| **PI Address 2:** | 204 Fenske Laboratory | | |
| **PI Web Page:** | | | |
| **City:** | University Park | **State:** | PA |
| **Zip Code:** | 16802 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-01 |
| **Start Date:** | 10/01/2001 | **End Date:** | 09/30/2006 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG9-1372 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | We propose to design and prototype biosensor array systems capable of optically detecting and identifying a wide variety of airborne trace organic, potentially toxic agents. These compounds include ketones, aldehydes, alcohols, alkanes, and halocarbons. This approach to the problem combines basic biological and materials science with novel microfabrication techniques. In specific we will immobilize fluorophore-labeled biorecognition molecules (apoenzymes) within novel poly(ethylene glycol) based gel materials that maintain protein conformation and thus biological function, and that can be patterned via photolithography or microcontact printing. These gels will be designed to allow partition of the agent from the gas phase into the gel where it can bind to the recognition molecule. To achieve optical detection, a fluorophore-modified ligand analogous to the agent of interest will be immobilized in the gel. In the absence of the agent, the ligand and recognition molecule will bind, resulting in fluorescence resonance energy transfer (FRET) between the respective fluorophores. When the agent is present, the ligand will be displaced competitively, FRET will cease, and the fluorescence in the gel increase. By micropatterning an array of these gels with different recognition | | |

| | |
|---|---|
| | molecules, multiple analytes can be simultaneously detected either directly or via pattern recognition techniques. This project will focus on the development of a fluorescent microsensor array for the following six model analytes: methanol, isopropanol, acetone, formaldehyde, chloroform, and toluene. Once patterned into an array via photolithography, the change in FRET upon exposure to gaseous analytes will be monitored using a CCD camera. The creation of such an arrays would represent a unique, specific, sensitive biologically-inspired method of gaseous agent detection in manned vehicles. Develop optical sensor arrays for volatile organic compounds, using biological recognition molecules to achiece recognition. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The identification and measurement of toxic volatile organic compounds and airborne pathogens are of critical concern not only to manned space flight, but to all Americans. There are numerous examples where sensors for these materials would be extremely useful. The bioterrorism attacks of 2001 were the result of exposure to an airborne pathogen, anthrax. Many rescue workers at the World Trade Center site suffer from respiratory problems, the result of prolonged exposure to toxic volatile organic compounds and smoke produced by the burning debris. Accounts of office workers becoming ill from toxins and pathogens in building ventilation systems ("sick building syndrome") have frequently occurred over the last two decades. Unfortunately, in all these cases, there are no rapid and accurate methods for quantifying airborne toxins and pathogens. This project seeks to create fast-response multianalyte sensors for these materials using biological inspired methods. |
| **Task Progress:** | Gel microstructures based upon poly(ethylene glycol) diacrylates, dimethacrylates, and tetraacrylates patterned photolithographically on silicon, glass, and plastic substrates were created and characterized. Substrates treated with 3-(trichlorosilyl)propyl methacrylate to form a self assembled monolayer (SAM) with pendant acrylate groups. SAM presence on the surface was verified using ellipsometry and time-of-flight secondary ion mass spectrometry. A solution containing an acrylated or methacrylated poly(ethylene glycol) derivative and a photoinitiator (2,2-dimethoxy-2-phenyl-acetophenone) was spin-coated onto the treated substrate, exposed to 365 nm ultraviolet light through a photomask and developed with either toluene or water. As a result of this process, three-dimensional, cross-linked PEG gel microstructures were immobilized on the surface. Diameters of cylindrical array members were varied from 600 um to 7 um by the use of different photomasks, while height varied from 3 to 12 um, depending on the molecular weight of the PEG macromer. In the case of 7 um diameter elements as many as 400 elements were reproducibly generated in a 1 mm2 square pattern. The resultant gel patterns were solvent-swollen for as long as three weeks without delamination from the substrate. In addition, micropatterning of different molecular weights of PEG was demonstrated. PEG gel microstructures were also created and arrayed using a microspotting robot, similar to that used to created DNA microarrays. Using this micropatterning technology, arrays of gels capable of detecting multiple analytes may be created without repeated lithographic patterning procedures. We demonstrated that these gel microstructures could be integrated with microfluidic systems for sample deliver via either a gas or liquid stream. Arrays of gel microstructures containing immobilized proteins conjugated to fluorophores were also prepared. Experiments were performed to verify that an enzyme may retain its activity in the gel microenvironment. A pH-sensitive fluorophore conjugated to an acid or base producing enzyme (e.g. acetylcholine esterase, urease) was encapsulated. The pH sensitivity of the gel-immobilized dye was similar to that in an aqueous buffer and no leaching of the dye-labeled protein from the hydrogel microstructure was observed over a one week period. Changes in fluorescence were also observed for immobilized fluorophore labeled acetylcholine esterase upon the addition of acetylcholine.<br>In addition, we have developed microfluidic devices that are capable of detecting airborne chemicals that have partitioned into an aqueous phase. Using a fluorophore-label antibody for the agent of interest, we were able to quantify the agent by the change in fluorescence anisotropy upon agent binding to the antibody. This sensing scheme was successful in detecting ppm quantities of styrene, benzene, and anthracene.<br><br>We finally developed an easy and effective method for the encapsulation of cells inside PEG-based hydrogel microstructures fabricated using photolithography and microreaction injection molding. High-density arrays of three-dimensional hydrogel microstructures were created on substrates using this method. Mammalian cells were encapsulated in cylindrical hydrogel microstructures of 600 and 50 um in diameter or in cubic hydrogel structures in microfluidic channels. Reducing lateral dimension of the individual hydrogel microstructure to 50 um allowed us to isolate 1 to 3 cells per microstructure. Viability assays demonstrated that cells remained viable inside these hydrogels after encapsulation for up to 14 days. Multiple pheotypes were also fabricated on a single substrate in independent microfluidic channels. These microstructures could be combined with a microfluidic device to create optical biosensor arrays of individually addressable single or multiple cell-containing hydrogel microstructures with potential applications in pathogen detection. |
| **Bibliography Type:** | Description: (Last Updated: 08/24/2007) |
| **Articles in Peer-reviewed Journals** | Koh, Won-Gun; Pishko, M. "Immobilization of multi-enzyme microreactors inside microfluidic devices" Science Direct , Aug-2004 |
| **Articles in Peer-reviewed Journals** | Yadavalli, V.; Russell, R.; McShane, M.; Cote', G.; Pishko, M. "A Monte Carlo Simulation of Photon Propagation in Fluorescent Poly(ethylene glycol) Hydrogel Microsensors" Science Direct , Aug-2004 |
| **Articles in Peer-reviewed Journals** | Yadavalli, V.; Pishko, M. "Biosensing in microfluidic channels using fluorescence polarization" Science Direct , Mar-2004 |
| **Articles in Peer-reviewed Journals** | Yadavalli, V.; Koh, W.-G.; Lazur, G.; Pishko, M. "Microfabricated protein-containing poly(ethylene glycol) hydrogel arrays for biosensing" Science Direct , Jan-2004 |